Sequential Causal Training of Windowed PINNs for the Lorenz ODE System

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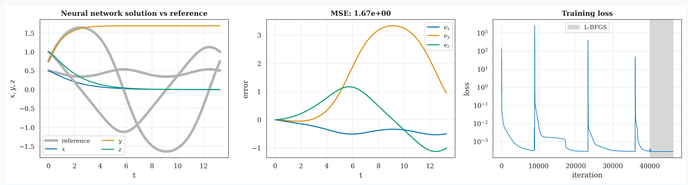
Final Year Design Project (FYDP-2) Defense

Code & saved run

The Long-Horizon Problem

One network per time window with causal weighting cuts the combined RMSE from 2.237 to 6.97×10^{-5}, trained only on the ODE residual.

- A full-interval PINN couples early and late times; later states depend on earlier ones.
- Errors accumulate and backtracking is impossible: our saved single-network run stalls at RMSE 2.237.
- Wang et al. (2024) fit early times first; we add time windowing.



Single network on the full interval. The trajectory drifts from the DOP853 reference; the loss plateaus. Saved-run RMSE: 2.237.

Lorenz-1960 IVP

GOVERNING EQUATIONS

$$\dot{x} = -0.1\,yz, \qquad \dot{y} = 1.6\,xz, \qquad \dot{z} = -0.75\,xy$$

Initial state u_0	(0.5, 0.75, 1.0)
Coefficients (k = 2, l = 1)	(−0.10, 1.60, −0.75)
Horizon	[0, 13.26446] · nearly closed orbit
Reference	DOP853 · rtol 10^{-10} · atol 10^{-12}
Evaluation	13,265 uniform times
Role of reference	monitoring only: never in the loss

- At the rounded endpoint the reference nearly returns to its start: a **nearly closed orbit**, not an exact period.

Windowed PINN

- Each window k has its own 4×60 tanh network: windows share boundaries, not parameters.
- A hard trial solution fixes every initial state **by construction**: no IC penalty, no weight to tune.
- Each window starts from the previous predicted endpoint; weights are copied as a warm start.

HARD TRIAL SOLUTION: EXACT IC

$$\hat{u}_k(t) = u_{0,k} + (t - t_k)\,N_k(t;\theta_k)$$

WINDOW HANDOFF + WARM START

$$u_{0,k+1} = \hat{u}_k(t_{k+1}), \qquad \theta_k \leftarrow \theta_{k-1}$$

Run Configuration

Interval / windows	[0, 13.26446] / 27 equal windows
Parameters	11,283 per window · 304,641 total
Collocation	1,536 uniform per window · 41,446 unique points
Optimizer	Adam 10^{-3} rate, ×0.9 every 1,000 steps
Causal stages	ε = 0.01, 0.1, 1, 10 · cap 4,000 steps
L-BFGS	≤ 1,000 closures per window
Precision / seed	float64 / seed 0

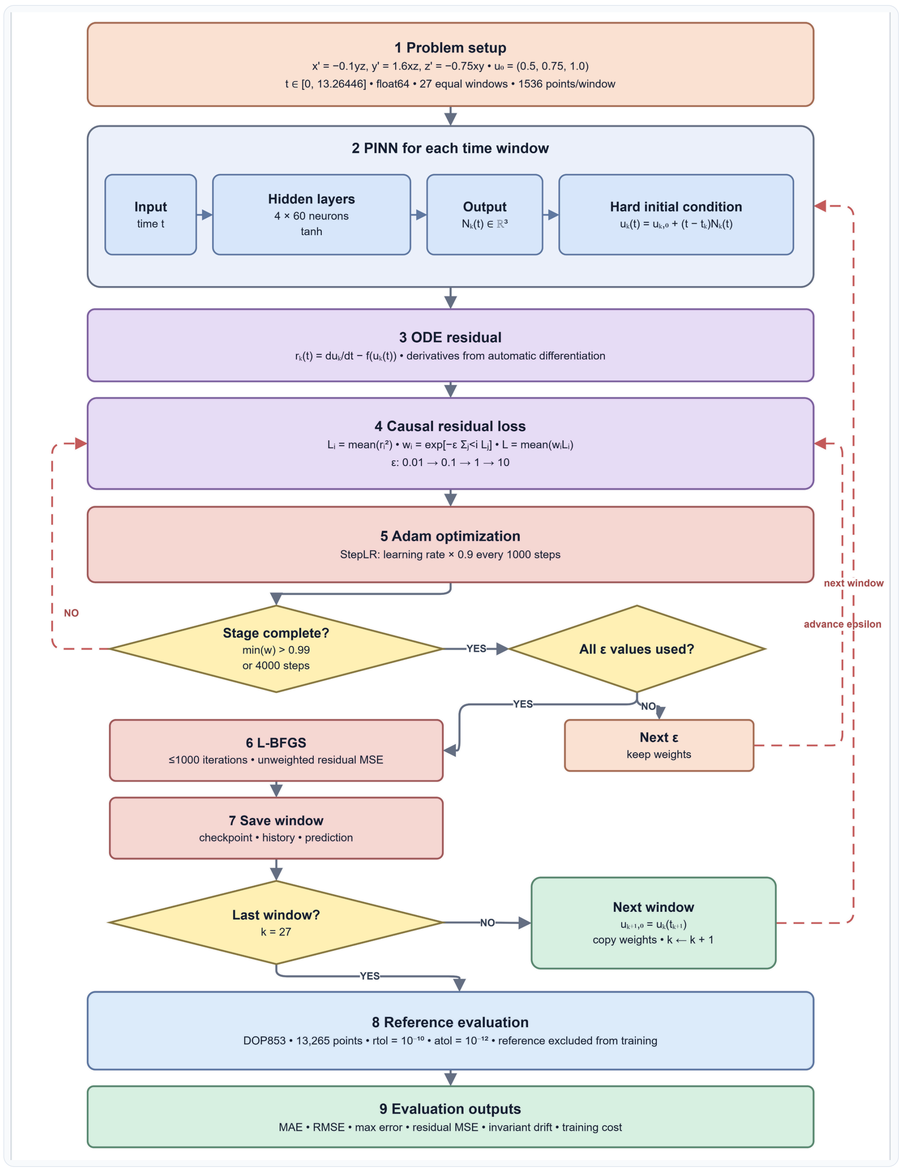
One Network vs 27 Windows

Measure	One network	27 windows
Trainable parameters	11,283	**304,641**
Training points	39,793 LHS	**41,446 uniform**
Combined state RMSE	2.237	**6.97×10^{-5}**
Max state-error norm	3.346	**3.09×10^{-4}**
Collocation residual MSE	2.80×10^{-4}	**1.87×10^{-7}**
Recorded wall time (s)	4,529	**881**

Descriptive comparison: several options change together, so no single factor is isolated.

34,765 Adam updates · ≤ 1,000 L-BFGS closures · 2 of 108 stages capped

Training Pipeline



Complete implementation pipeline. Per-window PINN with hard IC, four causal ε stages, L-BFGS polish, then handoff. The DOP853 reference never enters training.

Causal Weighting

- Residual losses are ordered in time: each weight depends on the accumulated loss of all earlier points, so the network must fit early times before late times.
- Weights are recomputed every epoch and stay fixed (detached) while Adam takes the gradient.

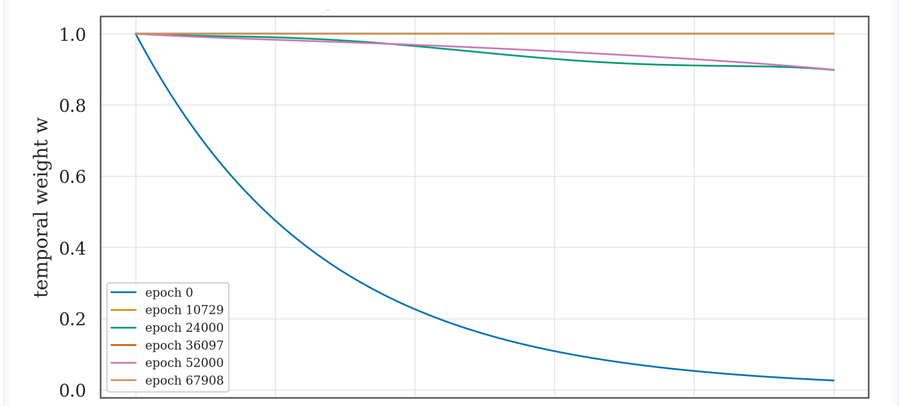
CAUSAL WEIGHT

$$w_{k,i} = \exp\Big(-\varepsilon \sum_{j<i} L_{k,j}\Big)$$

CAUSAL LOSS

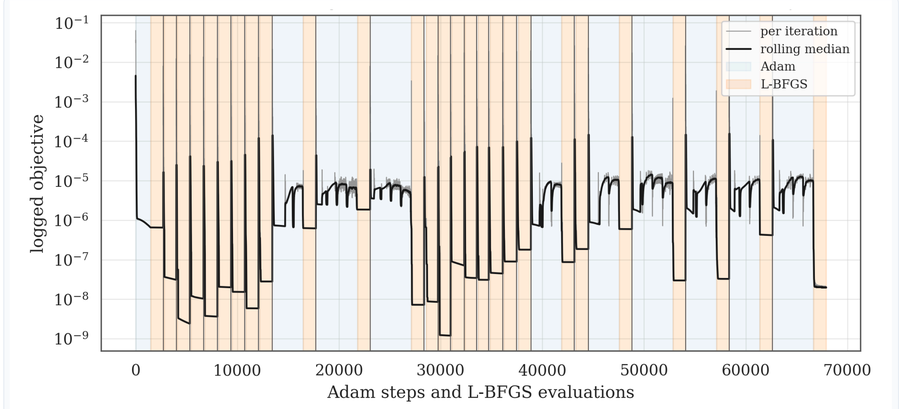
$$\mathcal{L}_{k,\varepsilon} = \frac{1}{n_k}\sum_{i=1}^{n_k} w_{k,i}\,L_{k,i}$$

advance when min w > 0.99 · then L-BFGS · unweighted residual mean



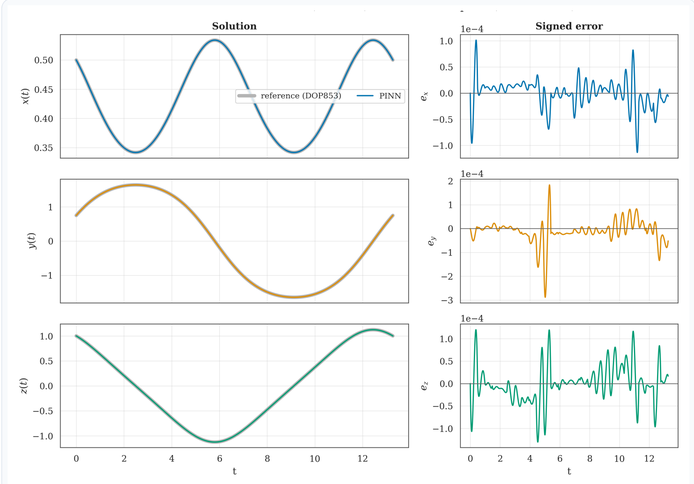
Causal weights at sampled training states. Only the window start is trusted early; weights flatten toward 1 as earlier losses shrink.

Optimizer Sequence



Adam and L-BFGS phases across all 27 windows. Adam uses causal weights; L-BFGS the raw residual mean. Each window's objective falls by orders of magnitude before the handoff.

Trajectory Accuracy



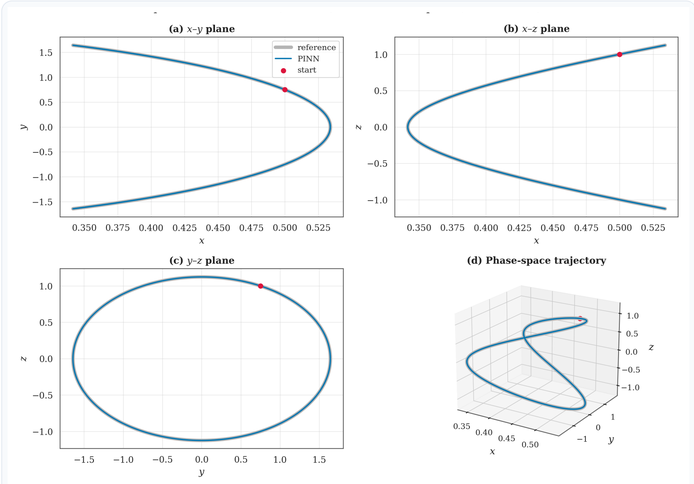
Windowed PINN vs DOP853 reference on 13,265 uniform evaluation times. Trajectories overlap; signed errors per state (×10^{-4}) peak near $t \approx 5$.

State	MAE	RMSE	Max
x	1.89×10^{-5}	2.74×10^{-5}	1.13×10^{-4}
y	2.94×10^{-5}	4.96×10^{-5}	2.88×10^{-4}
z	2.78×10^{-5}	4.05×10^{-5}	1.30×10^{-4}
State norm	**5.12×10^{-5}**	**6.97×10^{-5}**	**3.09×10^{-4}**

Physics Checks

- Residual MSE 1.874×10^{-7} (collocation) vs 1.872×10^{-7} (evaluation): generalisation gap ≈ 1.000.
- State error and ODE residual stay in the 10^{-5}–10^{-3} band along the whole orbit.
- Values are exactly continuous at all 26 inner interfaces, by construction of the hard trial solution.
- The DOP853 reference itself drifted less than 4×10^{-10}.

Phase Portraits



Phase portraits of the learned trajectory. The PINN (blue) tracks the reference (grey); the red dot marks u_0 = (0.5, 0.75, 1.0).

Limitations & Future Work

- Each window after the first starts from a **predicted** state: earlier error is carried forward, and derivatives can still jump at the interfaces.
- One seed, one initial state, one coefficient set. The two-run comparison changes several options at once, and no windowed run without causal weights was saved.
- Next: one-change-at-a-time runs with several seeds, and problems with boundary conditions at each end.

Conclusion

The 27-window causal PINN tracks the nearly closed Lorenz-1960 orbit at a combined state RMSE of 6.97×10^{-5} against DOP853; a single full-interval network recorded 2.237.

Trained from the ODE residual alone; the reference is used only for evaluation. Code, config and saved run: github.com/ihmorol/lorenz1960-pinn

Lorenz (196) Tellus · Raissi, Perdikaris & Karniadakis (2019) J. Comput. Phys. · Lagaris, Likas & Fotiadis (1998) IEEE Trans. Neural Netw. · Wang, Sankaran & Perdikaris (2024) Comput. Methods Appl. Mech. Eng. · Penwarden et al. (2023) J. Comput. Phys. · Matthews & Bihlo (2024) PinnDE · Virtanen et al. (2020) SciPy 1.0